

JEE MAIN REVISION MASTER: THREE DIMENSIONAL GEOMETRY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 41 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Planes in 3D & Line-Plane Intersections	23 PYQs	Planes
Very High	Direction Cosines, Direction Ratios & Lines in 3D	19 PYQs	3D Geometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Planes in 3D & Line-Plane Intersections	$d = \frac{ ax_1 + by_1 + cz_1 + d }{\sqrt{a^2 + b^2 + c^2}}, \quad \text{Line} \parallel \text{Plane} \iff \vec{b} \cdot \vec{n} = 0, \quad \text{Line} \perp \text{Plane} \iff \vec{b} \parallel \vec{n}$
Direction Cosines, Direction Ratios & Lines in 3D	$l^2 + m^2 + n^2 = 1, \quad a_1 a_2 + b_1 b_2 + c_1 c_2 = 0 \iff L_1 \perp L_2, \quad d_{\text{skew}} = \frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Planes in 3D & Line-Plane Intersections (Very High Yield — 23 PYQs)

Core Principle: General equation of plane: $ax + by + cz + d = 0$ where (a, b, c) are direction ratios of normal to plane. Perpendicular distance of point (x_1, y_1, z_1) from plane: $d = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$. Angle θ between line with direction $\vec{b}$ and plane with normal $\vec{n}$: $\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$.

■■ **Caution / JEE Pitfall:** Angle between LINE and PLANE uses SINE ($\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$), NOT cosine, because the angle is measured from the plane surface, not the normal!

■ **Speed Trick:** Two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ have perpendicular distance $d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The equation of the line through the point $(0,1,2)$ and perpendicular to the line $(x-1)/2 = (y+1)/3 = (z-1)/-2$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The distance of the point $P(-3,4,5)$ from yz plane is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The sine of the angle between the straight line $(x-2)/3 = (y-3)/4 = (z-4)/-5$ and the plane $2x-2y+z=5$ is..." [Ans: N/A]

3.2 Direction Cosines, Direction Ratios & Lines in 3D (Very High Yield — 19 PYQs)

Core Principle: Direction cosines (l, m, n): $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ and $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$. Two lines with direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) are perpendicular if $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$; parallel if $(a_1 / a_2) = (b_1 / b_2) = (c_1 / c_2)$. Shortest distance between skew lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$: $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$.

■ **Caution / JEE Pitfall:** Direction cosines must satisfy $l^2 + m^2 + n^2 = 1$, whereas direction ratios (a, b, c) can be any proportional numbers ($l = \frac{a}{\sqrt{a^2+b^2+c^2}}$).

■ **Speed Trick:** Lines intersect if and only if the shortest distance between them is zero: $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "The measure of the angle between the lines $x=k+1, y=2k-1, z=2k+3, k \in \mathbb{R}$ and $(x-1)/2 = (y-2)/1 = (z-3)/1$ is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "The angle between the lines whose direction ratios are a, b, c and $b-c, c-a, a-b$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is

- (A) 1 or 4
- (B) -1 or 4 [Correct]
- (C) 1 or -4
- (D) -1 or -4

Correct Answer: B

Solution Analysis: Since the triangle is right-angled at B, the two sides meeting at B must be perpendicular. So, vectors $\vec{BA}$ and $\vec{BC}$ must satisfy $\vec{BA} \cdot \vec{BC} = 0$. Now, $A(2,4,3), B(4,a,9), C(10,-1,7)$. So, $\vec{BA} = A - B = (2-4, 4-a, 3-9) = (-2, 4-a, -6)$. Also, $\vec{BC} = C - B = (10-4, -1-a, 7-9) = (6, -1-a, -2)$. Since they are perpendicular, $(-2)(6) + (4-a)(-1-a) + (-6)(-2) = 0$. $-12 + (4-a)(-1-a) + 12 = 0$. $(4-a)(-1-a) = 0$. So, $4-a=0$ or $-1-a=0$. Hence, $a=4$ or $a=-1$. So the correct option is Option B-1 or 4.

Benchmark Problem 2: KCET 2025 (Q#1)

The equation of the line through the point $(0,1,2)$ and perpendicular to the line $(x-1)/2 = (y+1)/3 = (z-1)/-2$ is

- (A) $(x/3) = (y-1)/4 = (z-2)/-3$
- (B) $(x/-3) = (y-1)/4 = (z-2)/3$
- (C) $(x/3) = (y-1)/4 = (z-2)/3$
- (D) $(x/3) = (y-1)/-4 = (z-2)/3$

Correct Answer: None

Solution Analysis: The given line has direction vector $v = (2, 3, -2)$. We seek a direction $d = (d_1, d_2, d_3)$ with $v \cdot d = 2d_1 + 3d_2 - 2d_3 = 0$. One convenient choice is $d_1 = -3, d_2 = 4, d_3 = 3$ since $2(-3) + 3(4) - 2(3) = -6 + 12 - 6 = 0$. Hence the required line through $(0,1,2)$ is $(x-0)/-3 = (y-1)/4 = (z-2)/3$, which is Option B.